Kilburn Trains Limited

Ashhab Imran

COMP23111

11/08/23

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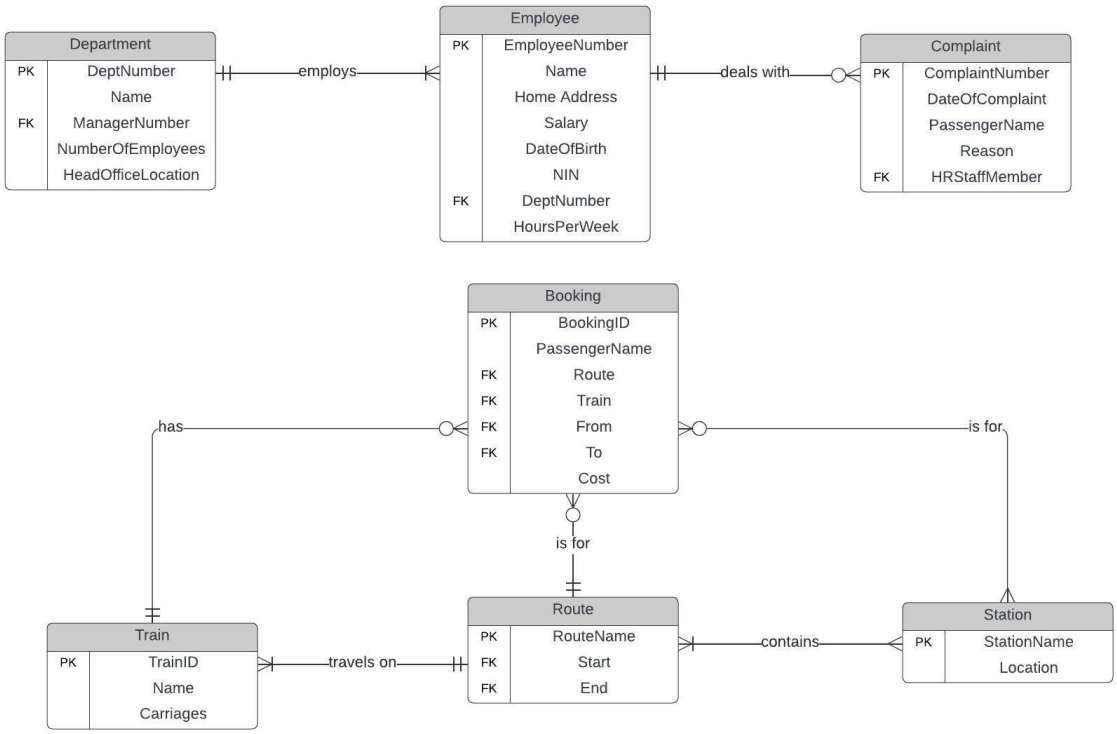
ERD

Here, we have an ERD diagram to outline the attributes and relations between each entity in the KTL database.

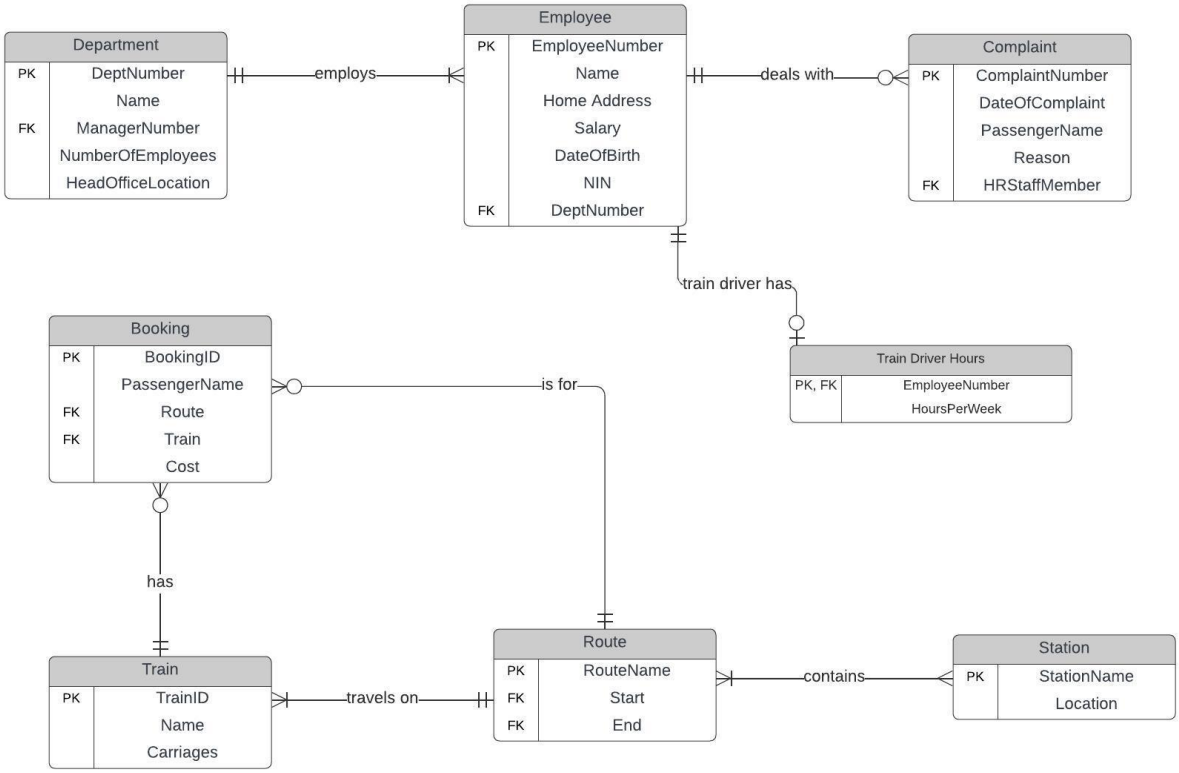


This diagram was primarily formed thanks to the provided material. The Department entity will contain four departments that make up KTL as a whole: Management, HR, Drivers, and Packagers. Each department will consist of a unique name, a number, an employee who manages it, several employees, and a head office location. Every employee will have a unique number, name, address, salary, date of birth, National Insurance Number, a department they are assigned to, and the hours per week for train drivers specifically. Each employee is assigned to one and only one department, and every department has several employees, including one manager. Each complaint will have a complaint number, the date of the complaint, the name of the passenger who made the complaint, and the reason. It is assigned to one and only one HR member, and every employee will deal with either no complaints or many complaints (HR members only). The booking entity has foreign keys from the train, route, and station entities. It is related to only one train and one route; however, it has multiple stations (at least 'from' and 'to'). Its remaining attributes consist of a booking ID, the passenger's name, and the ticket cost. Every train has an ID, a name, and a certain number of

carriages. It exists on one and only one route. Every route must have at least one train using it. This was not specified in the material, but from a logical point of view, it would be highly inefficient for every route to only have one train that uses it. The route's attributes are its unique name, starting station, and ending stations. The station entity consists only of the unique name and location. It has a many-to-many relation with the station entity.

Normalisation

I normalise my design to 3NF as best as I can. The following diagram is my ERD.

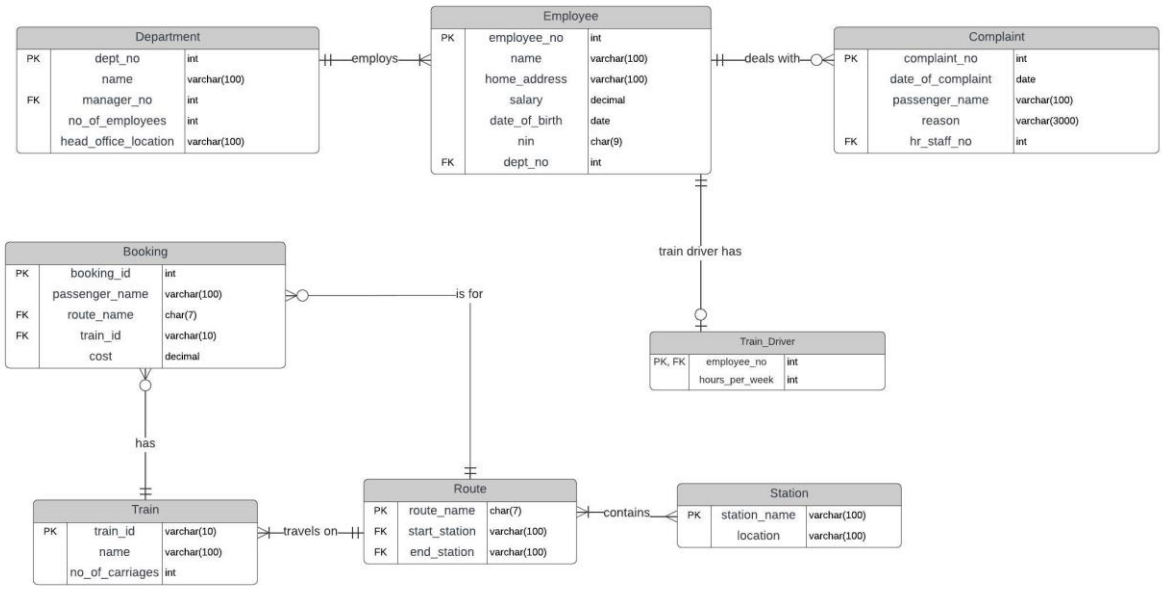


My previous ERD was already at 1NF: each entity had a primary key and there were no repeating groups. To normalise my diagram to 2NF, I added an additional entity titled 'Train Driver'. According to the material provided, we are supposed to keep track of the number of hours train drivers work a week. Train drivers' records are already in the 'employee' table, so I felt it was best to add another table for this to be recorded, because not every employee is a train driver, so the 'hours per week' column would be of no use to them. At the 3NF stage, I adjusted my 'booking' entity by removing the 'from' and 'to' columns entirely. The reason for this is because 'booking' already has a 'route' foreign key, which you can trace to the 'route' entity, and this already has the 'from' and 'to' attributes. This would make also having 'from' and 'to' in 'booking' pointless, because you can just

use the ‘route’ foreign key to get to the primary key in the ‘route’ table, which give you a ‘from’ and ‘to’. This meant I had to remove the relation between ‘booking’ and ‘station’ too.

Physical Data Model

To convert my ERD into a physical data model, I simply added data types for every attribute. I also made names clearer and more consistent, along with shortening some of them.



I shortened every attribute with the word ‘Number’ in it to ‘No’. I understand this may seem like a small and pointless change, but a shortened, more concise name is better to go by than longer names. In ‘route’, I changed ‘start’ and ‘end’ to ‘start_station’ and ‘end_station’ to be more specific because it was quite vague before to just be ‘start’ and ‘stop’. For similar reasons, I changed ‘carriages’ to ‘no_of_carriages’. Again, I did this because it was too unclear before that it specifies the number of carriages. I also changed the attribute names, so they use an underscore to conform to the appropriate rules of a physical data model. I also felt changing ‘train driver hours’ to just ‘train_driver’ was better and more concise. To convert the rest of the ERD, it was just a matter of adding data types to each attribute. For the ‘department’ entity, the department number, manager number, and number of employees are all INT, and the name and head office location have VARCHAR with a maximum of 100. In ‘employee’, the name and home address have a maximum VARCHAR of 100, but the national insurance number is CHAR with a size of 9. Typically, a national insurance number follows a specific format: two letters followed by six digits and ending with a single letter. The employee number and department number are both INT (which fits with the department’s department number). The salary is in DECIMAL and the date of birth is, of course, DATE. For ‘complaint’, the complaint number and HR staff number are INT, the date of complaint

and passenger name should be self-explanatory, but the reason has a limit of 3000 to allow for a detailed explanation for passenger complaints. As for the remaining entities, their data types are specified above. But for the ‘route name’ in route, the reason it has a size of 7 is because the example given in the material was in the following format: red-002. Though it is not shown in the diagram, every attribute except ‘hr_staff_no’ in the ‘complaint’ table is NOT NULL because it’s likely the complaint hasn’t been assigned to an employee yet.